학생용 종합과정 실험 A 자료

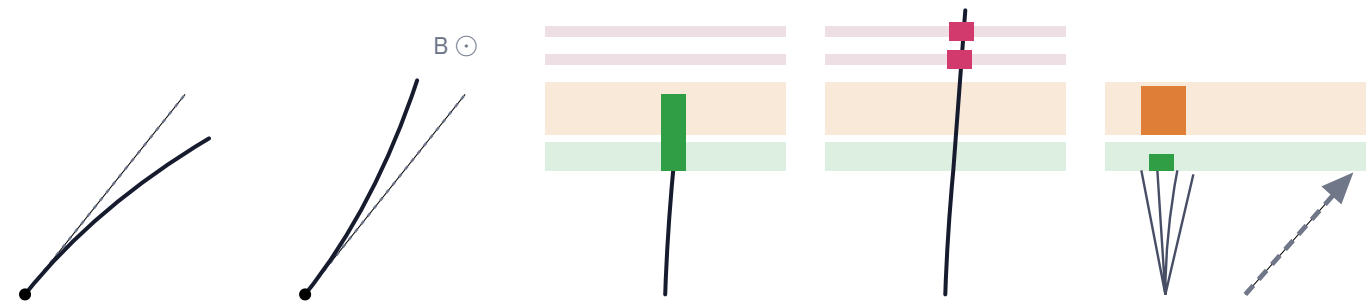
주제 9 · 사건 카드

사건 카드 40장: 세트 A

CMS와 비슷한 원통형 검출기를 빔 방향에서 본 그림(횡단면)입니다. 카드마다 신호 지문을 읽고 W^+, W^-, Z, 배경 가운데 어디에 속하는지 분류하세요.

모둠마다 서로 다른 세트(A~F)를 받으면 학급 전체가 서로 다른 사건 240개를 분석하게 됩니다. 세트는 언제 열어도 같은 카드로 만들어집니다. 인쇄는 A4 세로, 한 줄에 3장씩 나옵니다.

읽는 법 카드 읽는 법



자기장 B는 종이 면에서 나오는 방향(⊙, 3.8 T)입니다. 점선은 "자기장이 없다면 곧게 갔을 길(직진 기준선)"입니다. 보기 쉽도록 트래커 안의 휘어짐을 크게 과장했습니다(실제 40 GeV 트랙은 반지름이 약 35 m라서 거의 직선).

분류 규칙

- **W → eν / μν**: p_T가 큰(20 GeV 이상) **고립된** 렙톤 1개 + **큰 MET**(20 GeV 이상). 렙톤의 전하가 곧 W의 전하.
- **Z → ee / μμ**: 같은 종류, **반대 전하** 렙톤 2개(각각 20 GeV 이상) + MET 작음. 두 렙톤 불변질량이 약 91 GeV.
- **배경**: 트랙 다발(제트)만 있거나, 렙톤이 제트 속에 묻혀 있고 MET가 작은 사건.

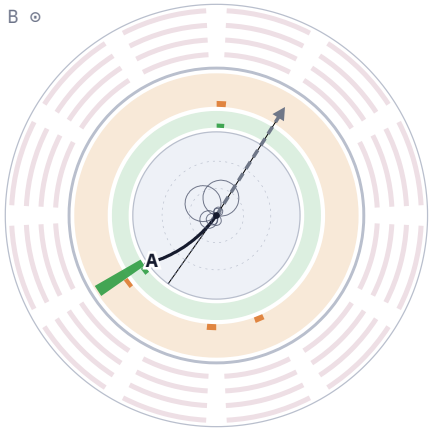
카드 숫자

- **트랙 A, B**: p_T 15 GeV가 넘는 트랙에만 이름을 붙였습니다. p_T는 횡운동량(GeV/c), η는 빔 방향과 이루는 각을 나타내는 값, φ는 충돌점에서 출발한 방향(오른쪽 = 0°, 반시계 방향으로 증가)입니다.
- **MET**: 누락 횡운동량의 크기와 방향. 화살표가 길수록 큽니다.
- 도전: Z 후보는 $M^2 = 2 \cdot p_{T1} \cdot p_{T2} \cdot (\cosh \Delta\eta - \cos \Delta\varphi)$로 불변질량을 계산해 보세요.

카드 사건 카드

A-01

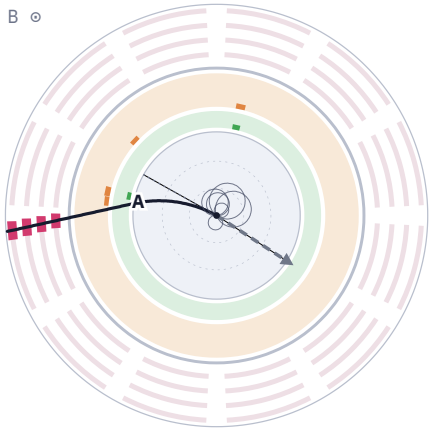
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	39.5	+0.07	235°
MET	39.5		58°

A-02

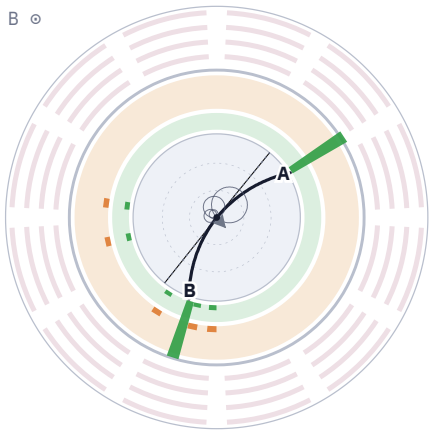
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	32.6	+0.13	151°
MET	27.5		327°

A-03

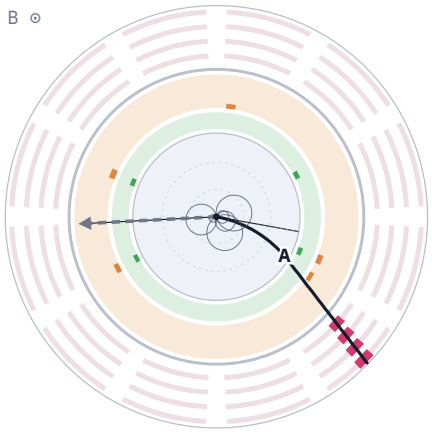
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	47.6	−0.31	51°
B	44.5	−0.06	231°
MET	1.9		311°

A-04

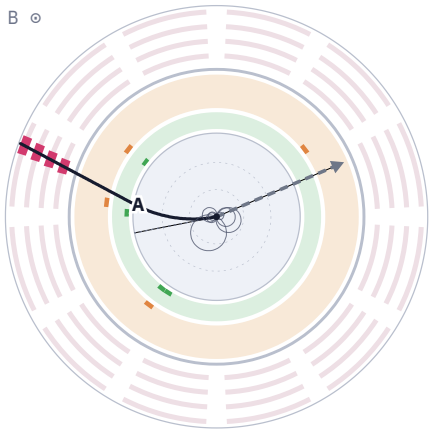
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	31.0	−0.23	350°
MET	42.8		183°

A-05

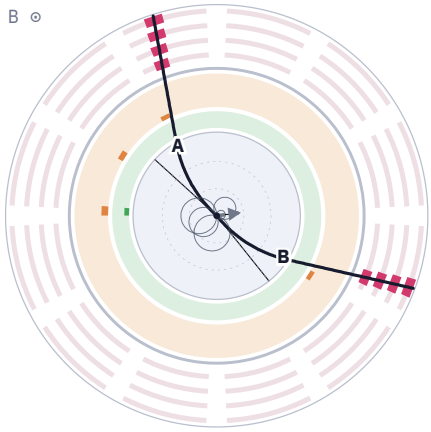
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	39.0	−0.21	191°
MET	43.0		23°

A-06

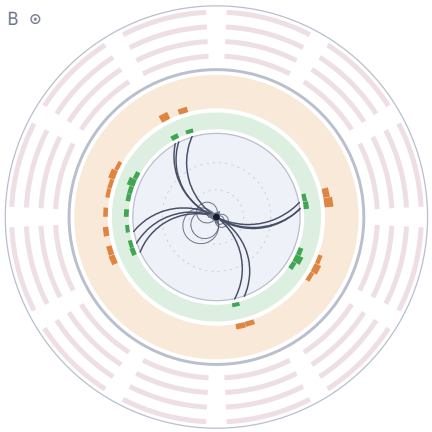
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	47.3	−0.41	138°
B	42.1	−0.52	309°
MET	5.3		7°

A-07

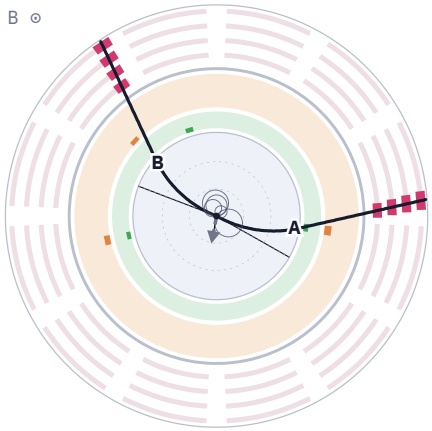
주제 9 · 세트 A



	p_T(GeV)	η	φ
	p_T 15 GeV 넘는 트랙 없음		
MET	0.5		170°

A-08

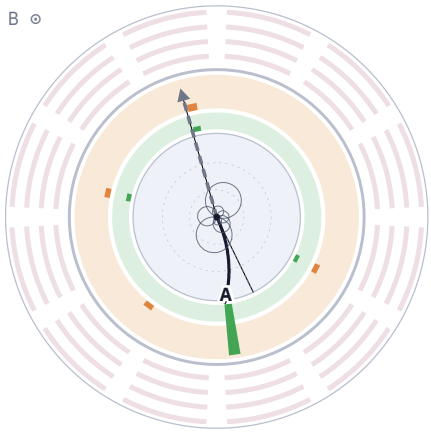
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	30.5	+1.06	330°
B	25.8	−1.07	159°
MET	6.5		259°

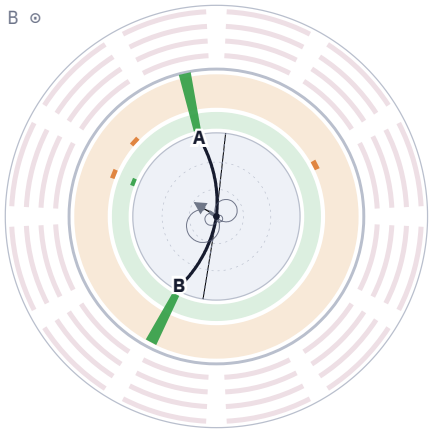
A-09

주제 9 · 세트 A



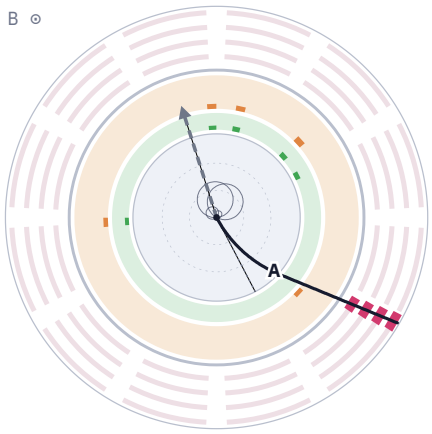
	p_T(GeV)	η	φ
A	38.2	+0.96	296°
MET	41.3		106°

A-10 주제 9 · 세트 A



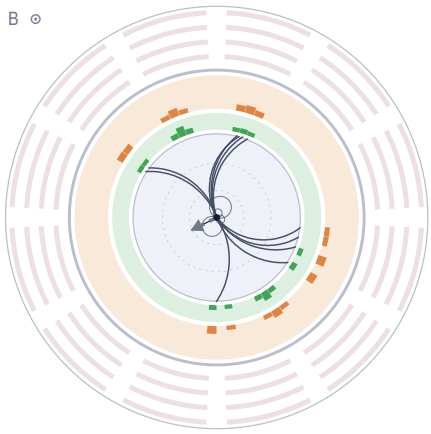
	p_T(GeV)	η	φ
A	43.7	−0.48	84°
B	40.6	+0.26	261°
MET	6.2		148°

A-11 주제 9 · 세트 A



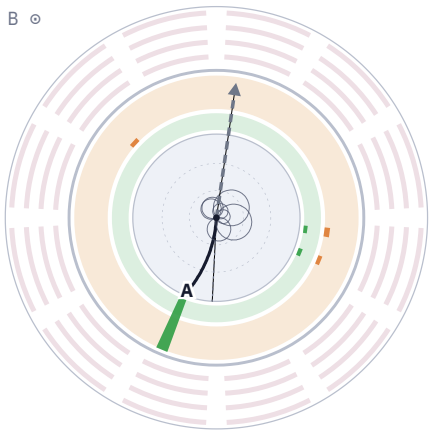
	p_T(GeV)	η	φ
A	38.0	+0.17	298°
MET	35.8		108°

A-12 주제 9 · 세트 A



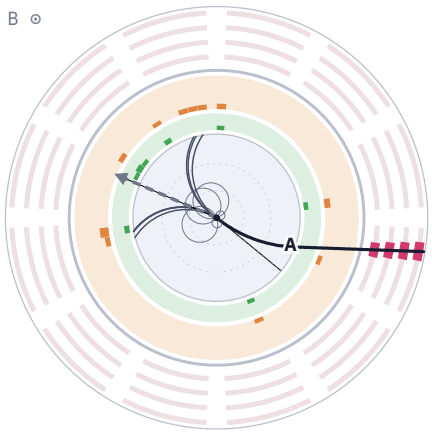
	p_T(GeV)	η	φ
	p_T 15 GeV 넘는 트랙 없음		
MET	7.0		207°

A-13 주제 9 · 세트 A



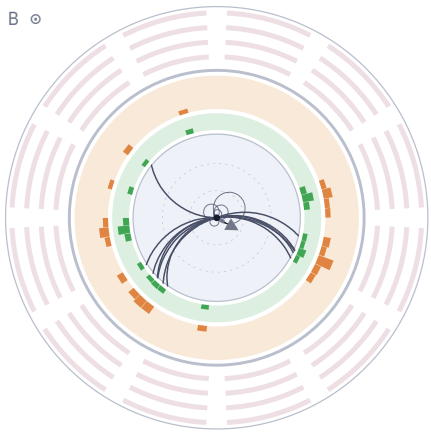
	p_T(GeV)	η	φ
A	41.4	+1.10	267°
MET	42.2		82°

A-14 주제 9 · 세트 A



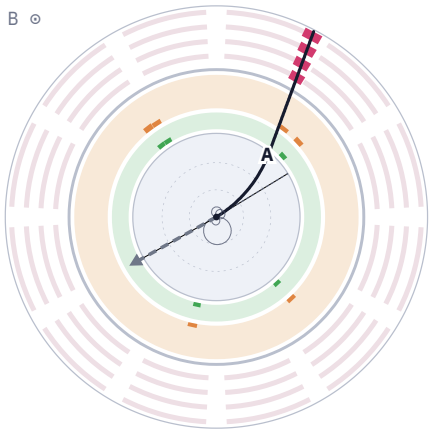
	p_T(GeV)	η	φ
A	47.7	−0.63	321°
MET	33.9		157°

A-15 주제 9 · 세트 A



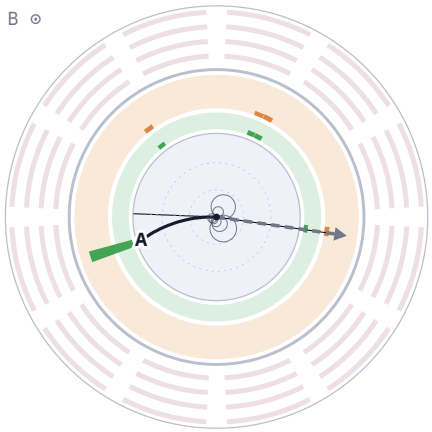
	p_T(GeV)	η	φ
	p_T 15 GeV 넘는 트랙 없음		
MET	5.5		330°

A-16 주제 9 · 세트 A



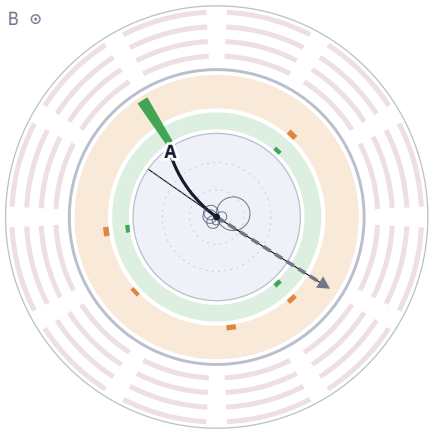
	p_T(GeV)	η	φ
A	41.8	−0.15	31°
MET	30.2		209°

A-17 주제 9 · 세트 A



	p_T(GeV)	η	φ
A	33.0	+0.25	178°
MET	40.7		352°

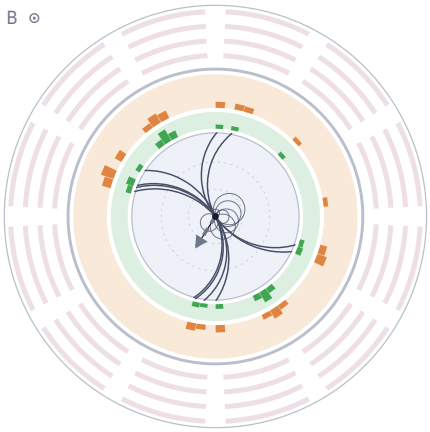
A-18 주제 9 · 세트 A



	p_T(GeV)	η	φ
A	37.8	+0.53	145°
MET	41.5		328°

A-19

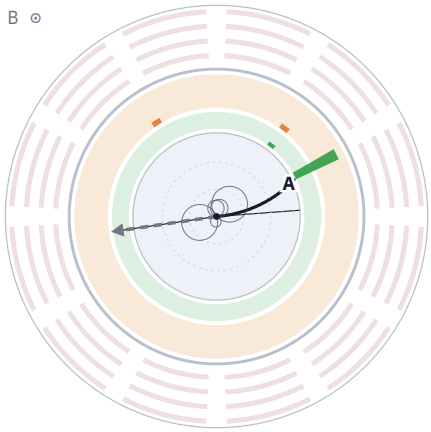
주제 9 · 세트 A



	p_T(GeV)	η	φ
	p_T 15 GeV 넘는 트랙 없음		
MET	9.8		238°

A-20

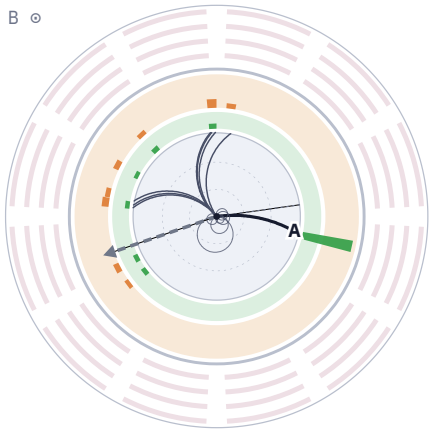
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	35.2	−1.11	4°
MET	32.4		188°

A-21

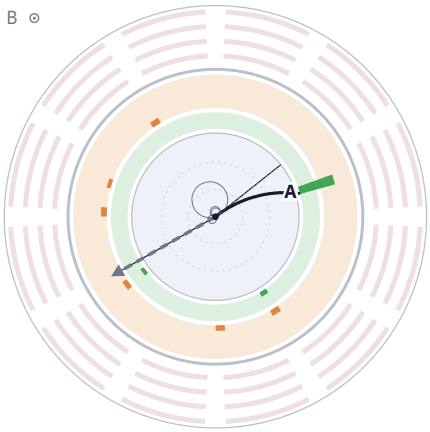
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	37.9	−0.53	8°
MET	36.9		199°

A-22

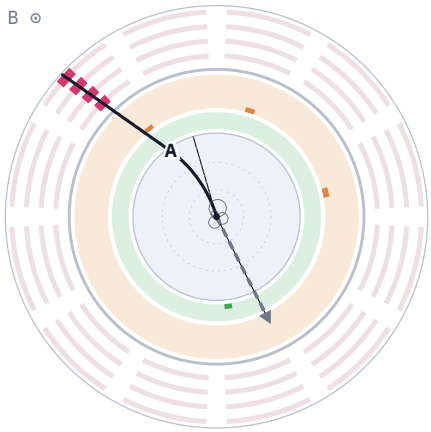
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	26.5	+0.63	38°
MET	36.9		210°

A-23

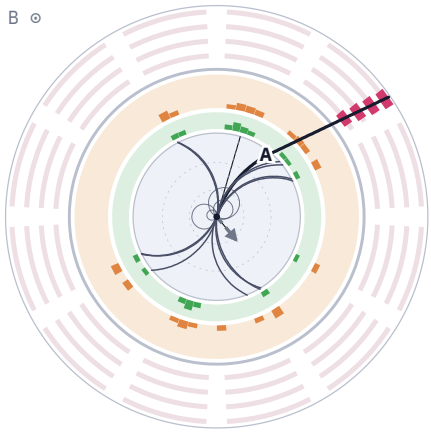
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	41.7	−1.13	106°
MET	36.9		297°

A-24

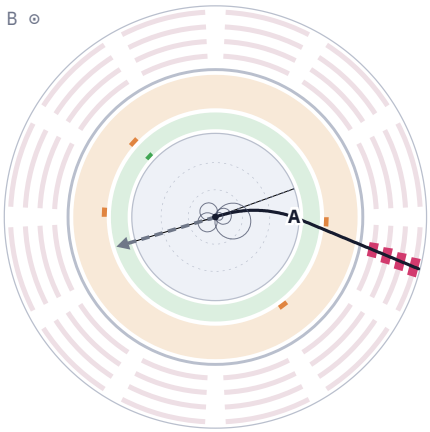
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	18.5	+1.25	74°
MET	8.0		310°

A-25

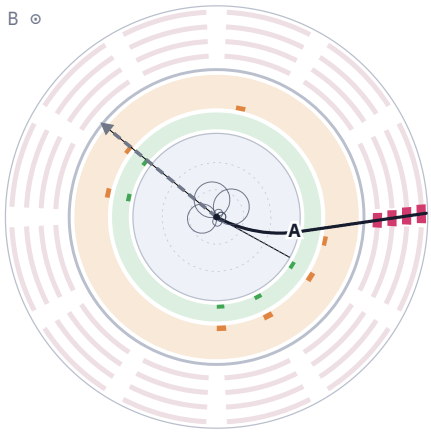
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	30.5	+0.12	20°
MET	31.4		197°

A-26

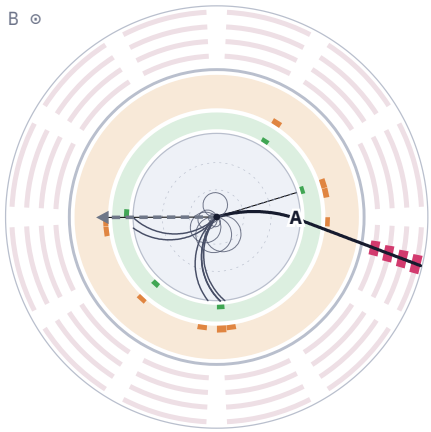
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	48.3	+0.48	331°
MET	46.7		141°

A-27

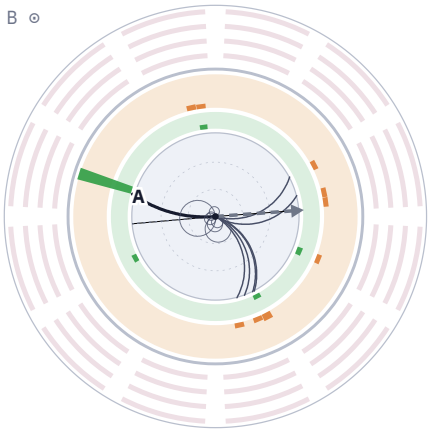
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	45.2	−0.98	17°
MET	37.0		180°

A-28

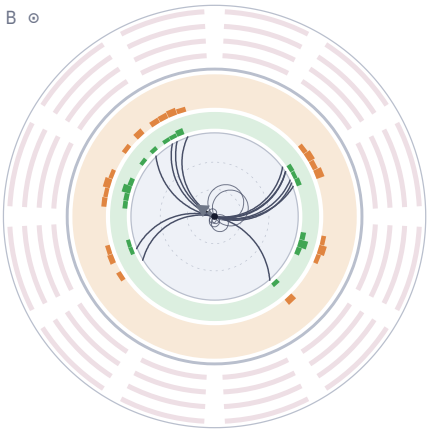
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	41.6	+0.78	185°
MET	26.5		4°

A-29

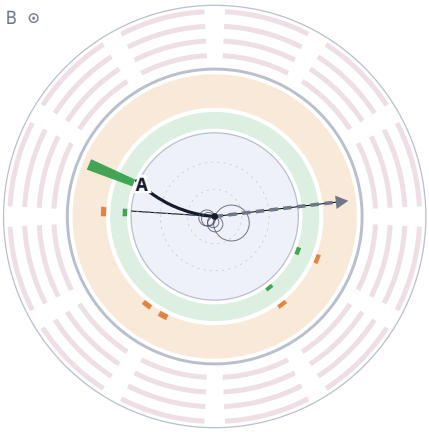
주제 9 · 세트 A



	p_T(GeV)	η	φ
	p_T 15 GeV 넘는 트랙 없음		
MET	4.5		148°

A-30

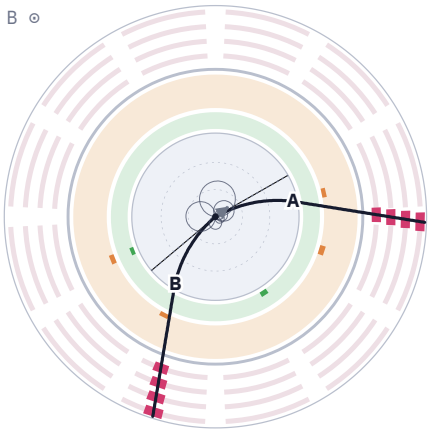
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	36.1	+0.31	176°
MET	41.6		7°

A-31

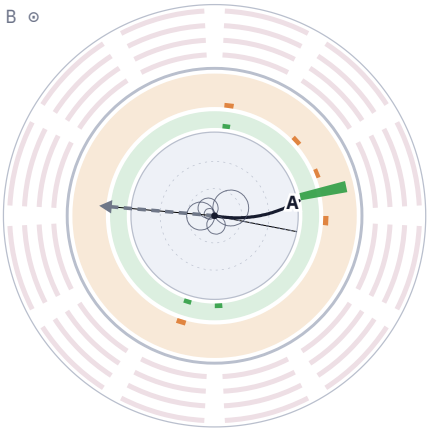
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	40.6	+1.04	30°
B	36.4	+0.06	220°
MET	2.8		37°

A-32

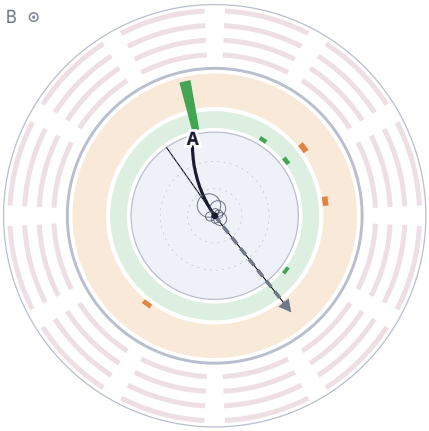
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	35.1	+0.13	349°
MET	35.5		175°

A-33

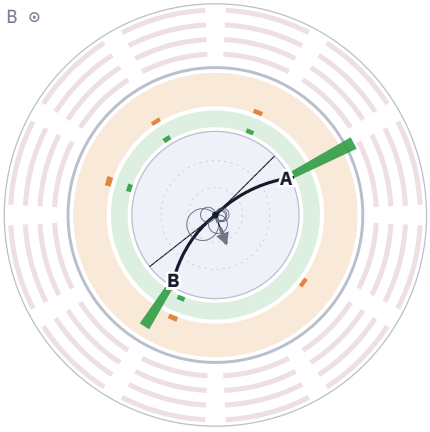
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	37.3	−0.16	125°
MET	37.9		308°

A-34

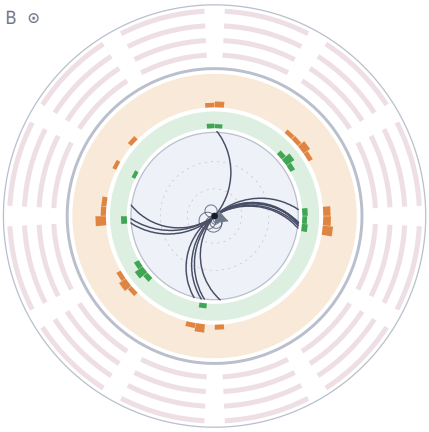
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	51.9	−0.76	45°
B	32.8	−0.49	218°
MET	8.0		291°

A-35

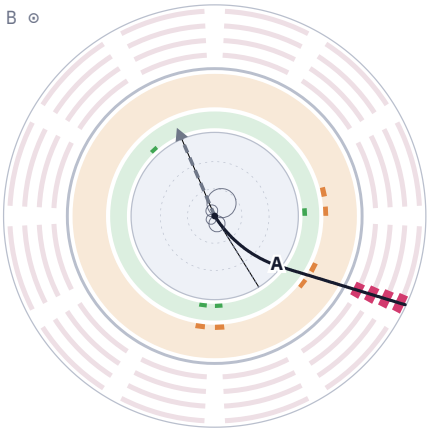
주제 9 · 세트 A



	p_T(GeV)	η	φ
	p_T 15 GeV 넘는 트랙 없음		
MET	2.1		340°

A-36

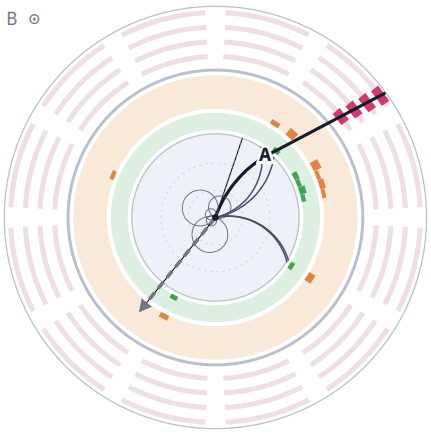
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	34.0	+0.73	302°
MET	28.9		113°

A-37

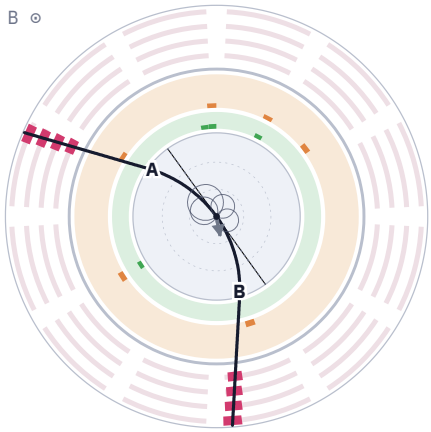
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	27.5	−0.10	71°
MET	37.5		231°

A-38

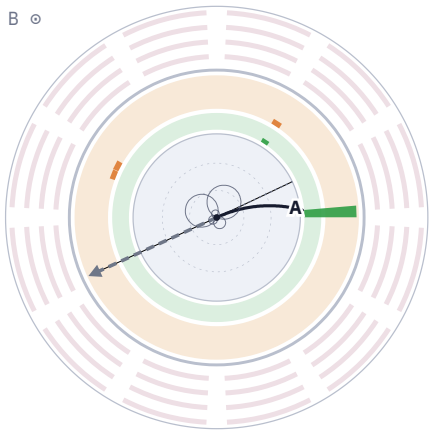
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	44.2	−0.02	126°
B	40.0	−0.80	306°
MET	4.1		281°

A-39

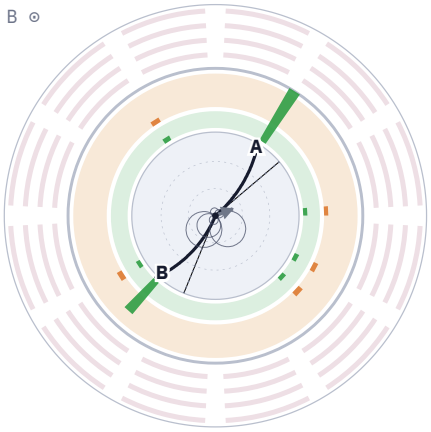
주제 9 · 세트 A



	p_T(GeV)	η	φ
A	39.1	+0.51	25°
MET	43.8		205°

A-40

주제 9 · 세트 A



	p_T(GeV)	η	φ
A	45.3	+0.74	40°
B	30.2	−0.55	248°
MET	3.8		21°

기록

모둠 분류 기록지

카드를 분류할 때마다 아래에서 골라 두면 집계가 자동으로 됩니다. 이 집계를 활동지의 집계표에 옮겨 적으세요. 종이로 할 때는 이 쪽을 인쇄해 빈칸에 적습니다.

A-01	선택	A-02	선택	A-03	선택
A-04	선택	A-05	선택	A-06	선택
A-07	선택	A-08	선택	A-09	선택
A-10	선택	A-11	선택	A-12	선택
A-13	선택	A-14	선택	A-15	선택
A-16	선택	A-17	선택	A-18	선택
A-19	선택	A-20	선택	A-21	선택
A-22	선택	A-23	선택	A-24	선택
A-25	선택	A-26	선택	A-27	선택
A-28	선택	A-29	선택	A-30	선택
A-31	선택	A-32	선택	A-33	선택
A-34	선택	A-35	선택	A-36	선택
A-37	선택	A-38	선택	A-39	선택
A-40	선택				

분류	개수 N	√N
$W^+ \to e^+\nu$	0	0.0
$W^- \to e^-\bar{\nu}$	0	0.0
$W^+ \to \mu^+\nu$	0	0.0
$W^- \to \mu^-\bar{\nu}$	0	0.0
$Z \to e^+e^-$	0	0.0
$Z \to \mu^+\mu^-$	0	0.0
배경	0	0.0
W^+ 합계 ($e^+\nu + \mu^+\nu$)	0	0.0
W^- 합계 ($e^-\bar{\nu} + \mu^-\bar{\nu}$)	0	0.0
Z 합계	0	0.0
분류한 카드	0	